Palaeontologia Electronica
palaeo-electronica.org

Rediscovery of the associated gigantic vertebrae of the extinct megatooth shark, *Otodus megalodon*, from the Upper Miocene Gram Formation in Denmark, and comments on its paleobiological significance and the maximum possible size of the species

Kenshu Shimada, Mette Elstrup, Henrik Lauridsen, Trine Sørensen, and Mikael Siversson

ABSTRACT

The extinct megatooth shark, *Otodus megalodon,* is an iconic Neogene shark that may have reached as much as 24.3 m in total length (TL). The 24.3 m TL estimate was partly based on published photographs of an associated set of vertebrae from the Upper Miocene Gram Formation in Gram, Denmark, some of which allegedly had a diameter of 23 cm. Here, we report on the rediscovery of the specimen, which was thought to be lost, confirming the reported maximum diameter of 23 cm. Those vertebrae co-occurred with remains of a cetorhinid (basking shark), but the morphology and size assessments of the vertebrae suggest that they belong to *O. megalodon*, with the possibility that the cetorhinid remains represent stomach content of the megatooth shark. The vertebral specimen of *O. megalodon* was also used to decipher its ontogenetic growth patterns, and this study suggests that this specific individual had the potential to grow even slightly larger than 24.3 m. However, the growth model presented in this work must be viewed as highly tentative. Nevertheless, the *O. megalodon* specimen described herein is significant because it represents the largest shark vertebral specimen known to date, and quite possibly even the largest non-tetrapod vertebrae ever recorded. Ultimately, a more complete skeletal specimen of *O. megalodon*, ideally with associated teeth, is required for any further discussion on the maximum possible size of the fossil species.

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Final citation: Shimada, Kenshu, Elstrup, Mette, Lauridsen, Henrik, Sørensen, Trine, and Siversson, Mikael. 2026. Rediscovery of the associated gigantic vertebrae of the extinct megatooth shark, *Otodus megalodon*, from the Upper Miocene Gram Formation in Denmark, and comments on its paleobiological significance and the maximum possible size of the species. Palaeontologia Electronica, 29(2):a27.
https://doi.org/10.26879/1674
palaeo-electronica.org/content/2026/5872-otodus-megalodon-vertebrae-from-denmark

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Keywords: development; Elasmobranchii; longevity; ontogeny; von Bertalanffy growth function

Submission: 21 February 2026. Acceptance: 14 May 2026.

INTRODUCTION

The megatooth shark, *Otodus megalodon* (Lamniformes: Otodontidae), is an iconic extinct shark whose fossil remains are commonly found in mid-Miocene–Early Pliocene marine deposits nearly worldwide (Cappetta, 2012; Pimiento and Balk, 2015; Boessenecker et al., 2019; Perez et al., 2021; Cooper et al., 2022; Shimada et al., 2022; Pollerspöck et al., 2023). Because no complete skeleton of the species is known to date, deciphering its biology, including simple metrics such as its total length (TL), has been challenging. Whereas the use of the extant white shark, *Carcharodon carcharias*, as a guide is now known to produce underestimated TL (Sternes et al., 2024), a recent study, that applied data of body part proportions in many extant and extinct shark taxa to two vertebral specimens of *O. megalodon*, suggested that the fossil shark could have reached as much as 24.3 m TL (Shimada et al., 2025).

One of the two vertebral specimens of *O. megalodon* used in Shimada et al.'s (2025) study, which was critical for the 24.3 m TL estimate, is an associated set of disarticulated gigantic centra reported from the Upper Miocene Gram Formation in Gram, Denmark, that purportedly measured as much as 23 cm in diameter (Bendix-Almgreen, 1982, 1983; Figure 1A–D). The vertebrae were excavated from Gram Claypit in 1978 and brought to the Geological Museum of Copenhagen (now part of the Natural History Museum of Denmark: NHMD), Copenhagen, Denmark. In relation to a move in 1989 from one storage facility to another, the specimen was severely damaged and was thought to have been lost, leaving behind only photographic records published by Bendix-Almgreen (1982, 1983) (e.g., Figure 1E, F). However, an attentive collection manager at NHMD recently rediscovered a small portion of the vertebral specimen, which is now formally cataloged as NHMD 157890.

NHMD 157890 is particularly noteworthy because it not only represents the largest vertebral specimen of *O. megalodon*, but, to our knowledge, also of any non-tetrapod vertebrate recorded to date. Besides the fact that body size plays a crucial role in every organism's life history, ecology, and evolution (e.g., Smith and Lyons, 2013, and references therein), documenting any exceptionally large fossil taxon is important from a paleobiological standpoint, particularly a gigantic carnivore like *O. megalodon*, that must have had a significant impact on its ecosystem (e.g., Shimada, 2019; Shimada et al., 2021a). Therefore, the purpose of this communication is to report the rediscovery of the materials represented by NHMD 157890, along with some new insights into the biology of *O. megalodon*.

GEOLOGICAL AND PALEONTOLOGICAL BACKGROUNDS

NHMD 157890 was discovered and collected in 1978 in a claypit in Gram, Denmark, that was dug into the Upper Miocene Gram Formation for brickworks (Figure 1A–D). The Gram Formation originally comprised three lithographic members: the Glauconite Clay, Gram Clay, and Gram Sand Members (Rasmussen, 1956). However, a recent stratigraphic revision (Rasmussen et al., 2010) restricted the Gram Formation to include only Rasmussen's (1956) Gram Clay Member, whereas the former underlying Glauconite Clay Member and overlying Gram Sand Member were elevated to formational level, the Ørnhøj Formation, and Marbæk Formation, respectively. The Gram Formation (sensu Rasmussen et al., 2010, which is used hereafter; Figure 1C, D) is a dark-brown clay unit with abundant siderite concretions in its lower part. In fact, many vertebrae of NHMD 157890 were found encased in such siderite concretions (Figure 1E, F). Based on the lithologic and paleontological evidence, the depositional setting of the formation, particularly the lower part, is interpreted to be a fully marine offshore environment >100 m deep (Laursen and Kristoffersen, 1999).

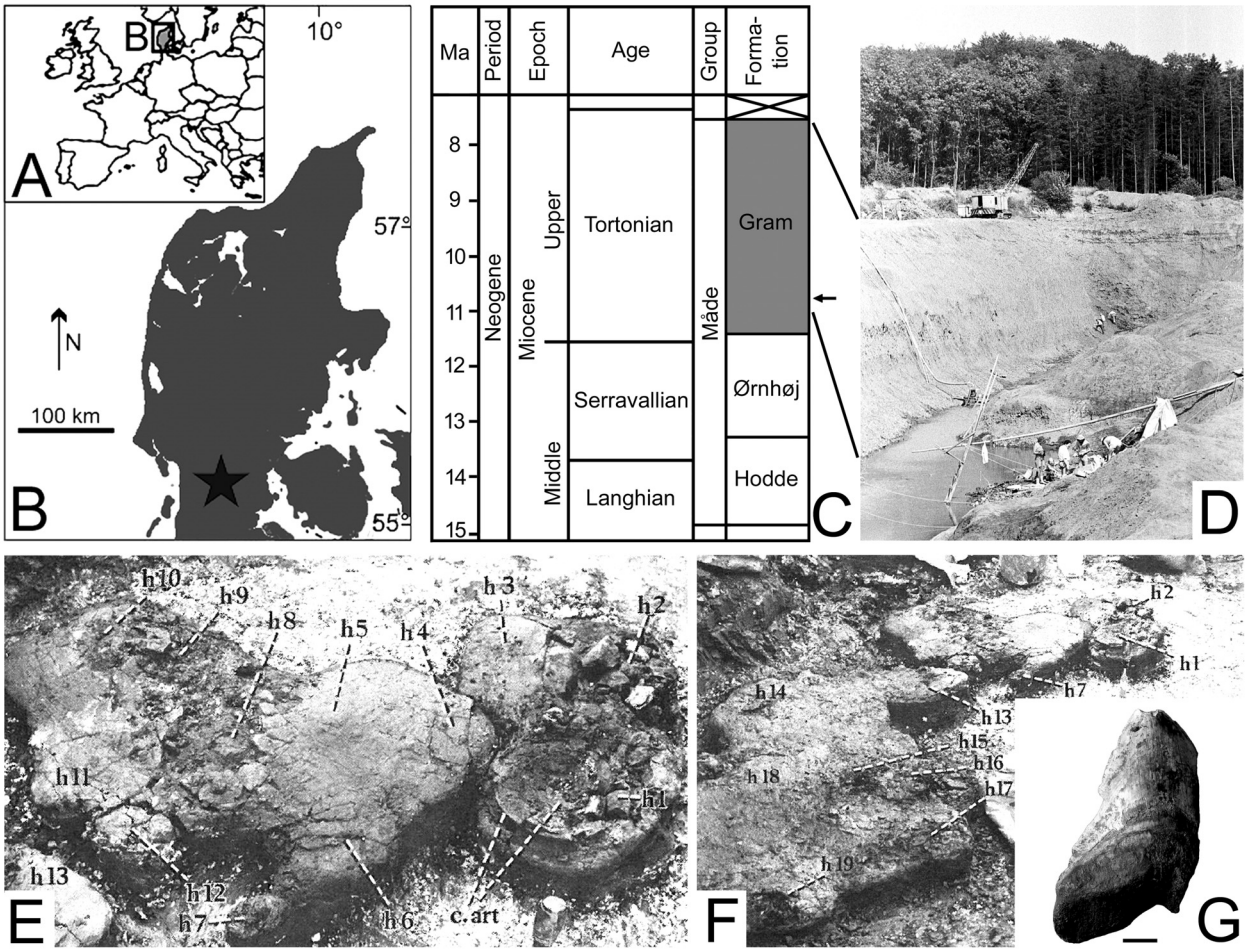


FIGURE 1. Geographic, geologic, and paleontological contexts of the associated gigantic shark vertebrae (NHMD 157890) described in this paper. **A**, Map of Europe, showing the position of Denmark (B). **B**, Map of part of Denmark showing the position of the fossil locality in Gram (star symbol). **C**, Middle–Upper Miocene stratigraphy of the region, highlighting the ca. 17-m-thick Upper Miocene Gram Formation in gray and the approximate stratigraphic horizon of NHMD 157890 pointed by an arrow, which is a few meters above the base of the formation (stratigraphy based on Rasmussen et al., 2010). **D**, Photograph of the locality showing the Gram Formation exposure and the excavation of NHMD 157890 in 1978. **E**, Close-up photograph of the excavation showing the distribution of associated vertebrae (NHMD 157890) presented by Bendix-Almgreen (1982, fig. 4D). **F**, Another photograph showing the distribution of associated vertebrae (NHMD 157890) presented by Bendix-Almgreen (1982, fig. 4C) (note that many vertebrae are encased in siderite concretions). **G**, Isolated tooth of *Otodus megalodon* (MSJN 1540; bar scale equals 2 cm) found at the same locality (see text for details). Image credits: D, courtesy of Gram Lokalhistorisk Arkiv (photograph taken by H. C. Gabelgaard); E and F, reproduced with permission from the editors of the now-discontinued journal *Nordslesvigske Museer*.

The exact thickness of the Gram Formation at the claypit where NHMD 157890 was found is uncertain because the mining operation did not reach the base of the formation, although the exposed section was nearly 15 m thick and capped largely by a Pleistocene glacial deposit (e.g., Rasmussen, 2005, fig. 5). However, a borehole sample taken near the claypit by the Geological Survey of Denmark in 1963 (DGU Borehole 141.277) showed that the Gram Formation at the locality is about 17.2 m thick. Because NHMD 157890 was found near the base of the claypit, its stratigraphic position is interpreted to be about 3–4 m above the presumed base of the formation (i.e., in the lower part of the formation: Figure 1C, D). This stratigraphic level represents the lower Tortonian, approximately 10.8 Ma (see Figure 1C, which is based largely on Rasmussen et al., 2010). Clay mining operation for brickworks was terminated in 1988, and the pit was altered in 1995 to accommodate public visits by reducing the cliff slope and pit depth for safety.

The Gram Formation is rich in fossils. Reported invertebrates comprise foraminiferans, dinoflagellates, bryozoans, echinoids, crustaceans, and diverse mollusks, including bivalves, scaphopods, gastropods, and cephalopods (e.g., Rasmussen, 1968; Kristoffersen, 1973; Laursen and Kristoffersen, 1999; Hansen and Hansen, 2003, 2005; Fraaije et al., 2005; Piasecki, 2005; Schnetler, 2005; Ramassamy, 2016). Remains of several fossil marine vertebrate groups have also been reported from the Gram Formation, including elasmobranchs (sharks and rays), actinopterygians (bony fishes), dermochelyoid sea turtle, alcid birds, and marine mammals, such as pinnipeds and cetaceans (e.g., Nordmann, 1944; Bendix-Almgreen, 1983; Hoch, 2004; Steeman, 2009; Karl et al., 2012; Koretsky et al., 2014; Gol'din and Steeman, 2015; Ramassamy, 2016; Ramassamy and Lauridsen, 2019; Reinecke et al., 2020; Schwarzhans and von der Hocht, 2023).

Besides NHMD 157890, the shark materials from the same claypit include an isolated tooth of *O. megalodon* reported by Bendix-Almgreen (1982, 1983), which was found in a sorting machine that processed mined clay coming from the same area at the time the associated vertebral specimen was discovered (MSJN 1540; Figure 1G; note that the species was assigned to the genus *Carcharodon* in his papers). Bendix-Almgreen (1982, 1983) thus considered the possibility that the tooth (MSJN 1540) and the associated gigantic vertebrae (NHMD 157890), which he also interpreted to belong to *O. megalodon*, may have come from the same shark individual. However, although we concur that the associated vertebrae do indeed belong to *O. megalodon* (see below), whether the tooth and vertebrae came from the same individual is questionable. Therefore, we conservatively treat these two specimens as separate entities, where this present work focuses on the associated vertebrae.

METHODS

Among many fragmentary vertebrae (see below), the two best preserved vertebrae in NHMD 157890, herein referred to arbitrarily as “Vertebra I” and “Vertebra II”, recently underwent extensive preparation by removing as much of the field jackets and matrix as possible. Vertebra I is still held in the original field jacket, and most of the siderite concretion that had formed around the vertebra has been removed. Vertebra II, which still accompanies a large amount of siderite concretion, was turned over in a new, lightweight jacket, and the original field jacket was subsequently removed completely. Sediment samples surrounding each vertebra were saved to examine the content of microscopic fossils.

The vertebral terminology used in this paper follows that of Newbrey et al. (2015). Besides external observations, a micro-computed tomography (microCT) scanner was also used on Vertebra I and Vertebra II. Vertebrae I and II were imaged with microCT (Vertebra I was scanned twice) using a TESCAN CoreTOM system equipped with an integrating detector and using the following parameters: Filter = Al 1.5 mm (Vertebra I, scans 1 and 2) or Cu 1 mm (Vertebra II), X-ray tube voltage = 170 kVp (Vertebra I, scans 1 and 2) or 160 kVp (Vertebra II), X-ray tube power = 92 W (Vertebra I, scan 1), 60 W (Vertebra I, scan 2) or 55 W (Vertebra II), integration time = 280 ms (Vertebra I, scan 1), 410 ms (Vertebra I, scan 2) or 180 ms (Vertebra II), isotropic spatial resolution = 0.092 mm (Vertebra I, scan 1), 0.06 mm (Vertebra I, scan 2) or 0.055 mm (Vertebra II), number of averages = 1 (Vertebra I, scan 1), 4 (Vertebra I, scan 2) or 3 (Vertebra II), acquisition mode = extended field-of-view (double acquisition) (Vertebra I, scan 1) or single field-of-view (Vertebra I, scan 2; Vertebra II), and acquisition time = ~8 h per sample. In addition, sediment samples were collected from Vertebrae I and II and other broken pieces of matrix and were sieved (mesh size: 0.1, 0.180, and 0.4 mm) to look for microscopic remains of fossil vertebrates.

Institutional abbreviations. IRSNB, Palaeontology Collection, Royal Belgian Institute of Natural Sciences, Brussels, Belgium; MCZ, Ichthyology Collection, Museum of Comparative Zoology, Harvard University, Cambridge, Massachusetts, USA; MNHC, Ichthyology Collection, Massachusetts Natural History Collection, University of Massachusetts, Amherst, Massachusetts, USA; MSJN, Museum of Southern Jutland, Natural History Collection, Gram, Denmark; NHMD, Natural History Museum of Denmark, Copenhagen, Denmark; ZMUC, Zoological Museum University of Copenhagen, now part of the Natural History Museum of Denmark, Copenhagen, Denmark.

SYSTEMATIC PALEONTOLOGY

Class CHONDRICHTHYES Huxley, 1880
Subclass ELASMOBRANCHII Bonaparte, 1838
Superorder SELACHIMORPHA Nelson, 1984
Order LAMNIFORMES Berg, 1958
Family OTODONTIDAE Glickman, 1964
Genus *OTODUS* Agassiz, 1838

Otodus megalodon (Agassiz, 1835)
Figures 1E, F, 2

Material. NHMD 157890, an associated vertebral specimen consisting of two partial centra (Vertebrae I and II) and at least 185 small vertebral fragments, including several rock pieces with natural casts of the vertebrae. The preserved vertebral parts are the remaining pieces of approximately 20 vertebrae originally reported by Bendix-Almgreen (1982, 1983).

Occurrence. Approximately 3–4 m above the base of the Gram Formation (Upper Miocene) of the Måde Group (sensu Rasmussen et al., 2010), Gram, Denmark (specifically from the lower Tortonian part, ca. 10.8 Ma; Figure 1A–D).

Description. Vertebrae I and II are partial centra that were preserved in plaster jackets, and they represent the best-preserved residual components of the original material reported by Bendix-Almgreen (1982, 1983). Vertebra I consists of about one-third of the articular side of one of the corpora calcarea, measuring up to about 5 mm in preserved thickness (Figure 2A, B). Although very incomplete, it preserves near the central portion of the centrum, allowing for reasonable estimation of the position of the focus (star symbol in Figure 2B), as well as three points representing its periphery, enabling measurement of its radius, which is 115 mm. The exact dorsoventral (or anteroposterior) orientation is uncertain, but, assuming a circular centrum shape, this radius translates into a diameter of 230 mm. The articular surface largely forms a shallow funnel shape and has a solid appearance. The innermost 17 mm area is slightly more steeply inclined relative to the rest of the funnel-shaped surface from which it is separated by the "angle change" (sensu Goldman et al., 2012; arrow mark in Figure 2A). Except for the innermost 17 mm, the articular surface of Vertebra I exhibits many concentric shallow grooves and low ridges, where each groove-ridge pair represents an incremental "growth band" (e.g., Cailliet and Goldman, 2004; Shimada et al., 2021b). Although some growth bands are less evident than others even in radiographs (Figure 2B, C), 28 outer growth bands are identifiable ("c"–"e": Figure 2B, C). The band intervals are relatively consistent between "c" and "d", with a range of 1.6–1.8 mm and an average of 1.706 mm, and both the intervals and prominence rapidly decrease in the last 12 bands, from 1.6 mm at "d" to 0.4 mm at "e." Parts of the corpus calcareum also exhibit a fine radial pattern externally (Figure 2A) and radiographically (bracket in Figure 2B), which may represent walls of "radial calcified tubes" (sensu Welton, 2013a, p. 22, fig. 13.1) or "radial canals" (sensu Siversson et al., 2019, p. 108, fig. 16c) present on the inside of the preserved corpus calcareum.

Vertebra II (Figure 2D–F) is also incomplete and exposes the inner surface of a partial corpus calcareum, forming a cone-shaped rise with many radially oriented ridges in the mid-section that may represent the walls of radial calcified tubes or canals (bracket in Figure 2E). Its funnel-shaped articular surface faces downward, and the siderite-based matrix fills the space between the articular surface and the new plaster jacket. The "summit" of the cone-shaped inner surface of the corpus calcareum is horizontally broken, exposing the siderite concretion in a circular manner, in which its center represents the focus of the circular vertebra. The corpus calcareum is only intermittently preserved and mostly still covered by matrix. It was not possible to determine a clear periphery, but the radius measured from the center of the siderite concretion to the outermost preserved parts of the corpus calcareum is 112 mm (Figure 2E). The sagittal(?) section made through the vertebra radiographically (Figure 2F) reveals that the thickness of the corpus calcareum near the focus of the vertebra is about 5 mm, the angle change occurs about 17 mm away from the vertebral focus, and the distance between the summit of the broken cone and the furthest extent of the level of the articular surface is about 35 mm.

NHMD 157890 also includes many vertebral fragments and some rock fragments preserving natural casts of small portions of the vertebrae. However, whether any of them belong to or come from Vertebrae I and II is uncertain, because they are too fragmentary. Anatomically, they are also less informative than what can be observed in Vertebrae I and II, although two noteworthy pieces are described here. Figure 2G shows a piece of siderite concretion preserving a natural cast of the central portion of a vertebral articular surface, and the "angle change" also occurs at about 17 mm away from the vertebral focus. Figure 2H is a vertebral fragment with 13 well-marked, rather tightly curved growth bands ranging from 0.9 to 1.6 mm in interval with an average of 1.108 mm. The outermost 6 mm periphery, which appears to mark the anatomical edge of the vertebra, has a blocky texture as a radial pattern of the corpus calcareum intersects with its concentrically oriented grooves, where each "block" appears to consist of many finely packed, unclear growth bands.

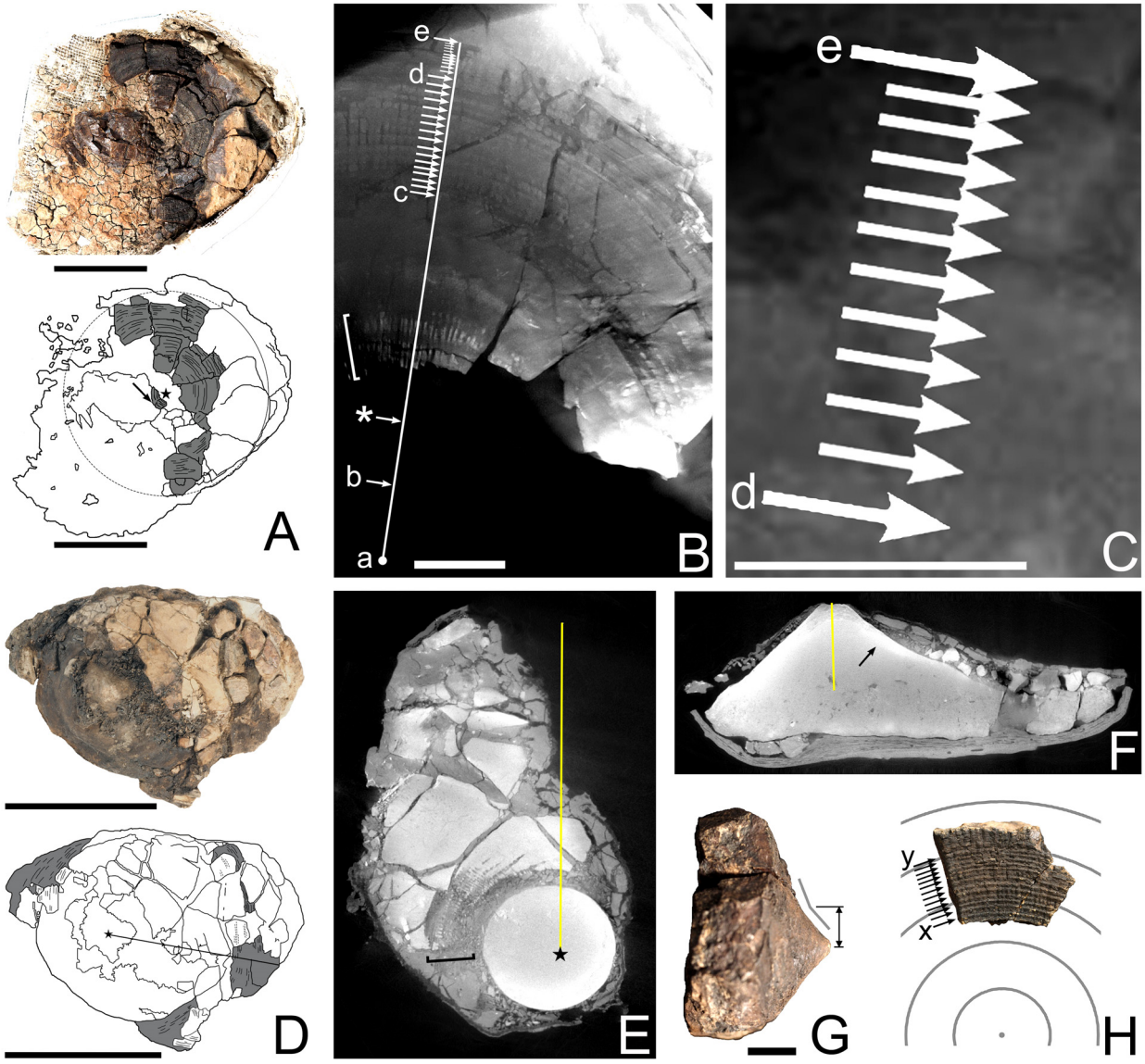


FIGURE 2. Associated vertebral specimen of *Otodus megalodon* (NHMD 157890) from the Gram Formation of Denmark. **A**, Photograph (top) and line drawing (bottom) of "Vertebra I", where the line drawing, which focuses only on parts exposing the exterior surface, shows the likely original outline (circle), focus (star), and position of "angle change" (arrow) of the vertebra (see text for details). **B**, Partial microCT image of Vertebra I, showing observed radial pattern (i.e., range of the bracket) and other specific anatomical positions discussed in the text. **C**, Close-up view of microCT image of Vertebra I at its periphery showing the last 12 interpreted growth bands from "d" to "e" shown in Figure 2B. **D**, Photograph (top) and line drawing (bottom) of "Vertebra II", where the line drawing shows the extent of the exposed, preserved vertebra (gray), its interpreted focus (star), and a line indicating vertebral radius (see text for details). **E**, MicroCT image of Vertebra II showing dense fractured siderite concretion along with a vertical line (star = focus; top end = periphery of the vertebra) indicating vertebral radius (= star and line in Figure 2D), bracket indicating the range of observed radial pattern. **F**, MicroCT image of Vertebra II showing sagittal(?) cross-section through the center of the vertebra, including the siderite concretion and plaster jacket, an arrow pointing the position of the "angle change" of the corpus calcareum, and a vertical line indicating the extent of one of the two "vertebral cones" used for the extrapolation of the total anteroposterior length of the vertebra. **G**, Rock matrix with natural cast of the center of a vertebral centrum in profile view showing the distance between the focus of the centrum (tip of the cone) and "angle change" (gray bent bar), which measures about 17 mm. **H**, Vertebral fragment that likely comes from a different vertebra than Vertebrae I or II, superimposed on concentric circles with an interval of 1 cm and showing growth bands (arrows) on its surface (see text for detail). Bar scale: A, D = 10 cm; B = 2 cm; C = 5 mm; E (vertical line) = 112 mm; F (vertical line) = 35 mm; G, H (each concentric interval) = 1 cm.

Remarks. Bendix-Almgreen (1982, 1983) noted that approximately 20 associated vertebrae were found deep in the claypit (Figure 1D). They were lying close to each other and were encapsulated within a couple of large, flat, irregularly shaped concretions. The vertebrae ranged from "about 10 cm to 23 cm" in diameter (Figure 1E, F). Bendix-Almgreen (1982) further noted that most of the centra were deformed due to compression of the clay. However, he added that a few of the larger ones were so well-preserved that the original shape could be determined with reasonable certainty, noting their anteroposterior lengths ("thicknesses") to be 5–8 cm and characterized them as short (or "thin").

Vertebra I (e.g., Figure 2A) described here is important because, with a radius of 115 mm, it substantiates the recorded maximum diameter of 230 mm. The periphery of Vertebra II is slightly damaged with a measurable radius of 112 mm (Figure 2E), which would give a vertebral diameter of at least 224 mm but likely no larger than 230 mm. Vertebra II is likewise important because it allows the inference about its anteroposterior length. The preserved half of one of its corpora calcarea measures 35 mm anteroposteriorly (Figure 2F), but because about 5 mm of the summit of the cone is estimated to be missing, one-half of the vertebra likely measured about 4 cm in length. If so, the original total anteroposterior thickness would have measured about 8 cm, which corroborates well with the upper limit of the vertebral thickness noted by Bendix-Almgreen (1982).

We also point out that the vertebral fragment illustrated in Figure 2H is significant because it is interpreted to have come from a vertebra with a 4.5 cm radius, or about 9 cm in diameter. Hence, this vertebra, which is clearly not from Vertebra I or II, may represent Bendix-Almgreen's (1983) smallest ("about 10 cm") measured vertebra. If so, NHMD 157890 is presently represented by at least three vertebrae. It should be added that this small vertebra is interpreted to have come from a much more caudal position than Vertebra I or II, not only because of its small size but also because of its tightly spaced growth bands (0.9–1.6 mm with an average of 1.108 mm) compared to at least Vertebra I (1.6–1.8 mm with an average of 1.706 mm in bands between "c" and "d"). The vertebral development is interpreted to have become significantly reduced after the formation of band "y" (Figure 2H), where its outermost growth bands are interpreted to have likely become juxtaposed and indistinguishable from one another. It is known that, depending on the species, not all growth bands may be deposited in shark vertebrae past a certain age, particularly in ontogenetically older sharks (e.g., Francis et al., 2007; Passerotti et al., 2014; Harry, 2018).

Associated fossils. Several pieces of siderite concretion collected from the same dig site and stratigraphic horizon as the vertebrae (NHMD 157890) were also found to be retained. Close examinations of the pieces reveal that some contain fragments of cetorhinid (basking shark: Lamniformes, Cetorhinidae) gill rakers (e.g., Figure 3A). Their filament widths range up to about 2 mm, and where preserved, the hook-like basal end of gill rakers varies in size and shape, with bight widths ranging from 2.5 mm to 5 mm and bight shape from "rounded" to "subangular" (terminology sensu Welton, 2013a, figs. 10, 11). However, the exact total number and morphology of the gill rakers are uncertain because they are too fragmentary. In addition, subsequent microscopic analysis of sediment samples from one of the softer rock pieces, which measured only about 200 cm^3, yielded more small fragments of cetorhinid gill rakers. More remarkably, a total of 75 "pieces" of cetorhinid placoid scales, three of which consist of two fused scales at their tooth roots, were also recovered from the sediment sample (four representative examples shown in Figure 3B). In addition, further inspection led to the recovery of additional cetorhinid gill raker fragments and placoid scales from sediments directly surrounding Vertebra II, substantiating their co-occurrence with the vertebral specimens. Each cetorhinid placoid scale, which measures 0.4–0.8 mm in maximum dimension, contains a posteriorly directed, narrow to robust, sharply pointed enameloid crown with faint, rounded ridges, that tend to converge apicoposteriorly. The crown sits on a broad-based, rough-surfaced dentin root with an irregularly jagged periphery and a flat to gently convex basal face that may bear one poorly defined nutritive pore.

Because of their high concentration, all gill raker fragments and placoid scales (e.g., Figure 3A, B) are regarded as originating from a single basking shark individual and are assigned a single catalog number, NHMD 1842036. However, the exact taxonomic identification of the gill rakers and placoid scales below the level of the family Cetorhinidae is difficult because the intraspecific and interspecific variations of these elements are still poorly understood. In particular, all the gill rakers are fragmentary, and the only informative description of cetorhinid placoid scales is that of Hovestadt

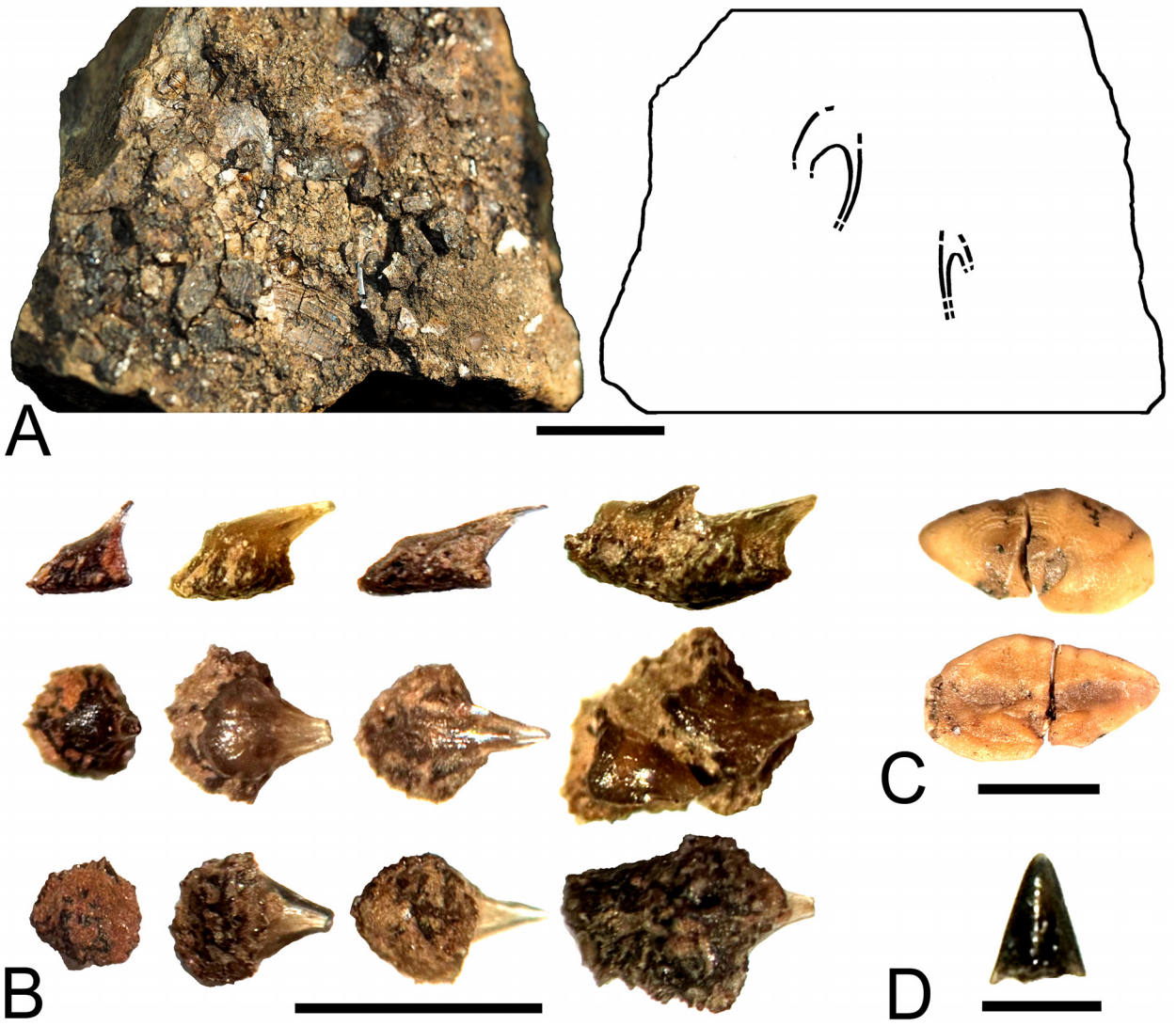


FIGURE 3. Remains of other fossil vertebrates that co-occurred with the vertebral specimen of *Otodus megalodon* (NHMD 157890) from the Gram Formation of Denmark. **A**, Photograph (left) and line drawing (right) of two likely associated gill rakers of a cetorhinid (basking shark: NHMD 1842036) in rock matrix. **B**, Four examples of placoid scales of a cetorhinid interpreted to belong to the same individual as NHMD 1842036 (all anterior to the left; top row = profile view; middle row = apical view; bottom row = basal view; note that the right-most specimen consists of two fused scales. **C**, Isolated otolith of gadiform teleostean fish, *Gadiculus deurnensis* Nolf (NHMD 2041713) in outer (top) and inner (bottom) views. **D**, Isolated tooth of an indeterminate teleostean fish (NHMD 2041714) in labial view. Bar scales: A = 1 cm; B, C = 1 mm; D = 0.2 mm.

and Hovestadt-Euler (2012), which was limited to those in the gill region of an Oligocene cetorhinid, *Keasius parvus* (Leriche), and the extant form, *Cetorhinus maximus* (Gunnerus). Those fossil and extant cetorhinid scales are morphologically similar to each other but quite different from those described here by exhibiting three prominent keels on the rather narrow apical crown face. It should be noted that these placoid scales are similar in size to the reported placoid scales of *O. megalodon* (0.3–0.8 mm) but are markedly different in morphology by lacking a major constriction (or "neck") between the crown and root as well as having converging rather than parallel ridges on the crown surface (see Shimada et al., 2024). The different morphology of the placoid scales in NHMD 1842036 described here (Figure 3B) indicates that they originate from a different part of the body; however, we do not rule out the possibility that it reflects differences at the taxonomic level rather than the positional level. Regardless, they are significant because they represent the second fossil

record of cetorhinid placoid scales to our knowledge.

Additionally, we note that, besides the presence of numerous microscopic remains of foraminiferans, ostracods, bryozoans, and oblong invertebrate(?)-derived coprolites, several isolated microscopic bony fish elements were also collected. They include a few unidentifiable bone fragments (uncatalogued), two otoliths from a gadiform teleostean fish, *Gadiculus deurnensis* Nolf (NHMD 2041712 and 2041713; one example shown in Figure 3C), and a triangular, spearhead-shaped tooth of an indeterminate taxon (NHMD 2041714; Figure 3D). However, unlike the cetorhinid remains represented by NHMD 1842036, these bony fish specimens are interpreted to represent isolated finds.

DISCUSSION

Taxonomic and Taphonomic Assessments

The species *"megalodon"* was formerly classified into various genera, such as *Carcharocles* Jordan and Hannibal, *Procarcharodon* Casier, *Megaselachus* Glickman, and even the extant white shark genus *Carcharodon* Smith (e.g., see Cappetta, 2012; Shimada et al., 2017). In fact, in the context of assessing the taxonomic placement of the associated gigantic vertebrae (i.e., NHMD 157890), Bendix-Almgreen (1982) also placed the species into *Carcharodon* by noting that the associated vertebrae are very similar to those of the recent white shark, both in proportion and calcification pattern, except for the significant difference in size. However, the species is now commonly placed in the genus *Otodus* (Otodontidae) to avoid direct phylogenetic linkage to *Carcharodon* (Lamnidae) and *Otodus* non-monophyly as well as to honor taxonomic priority (Ehret et al., 2012; Shimada et al., 2017, and references therein). This is because of the well-demonstrated morphological transition of the teeth from the non-serrated Miocene *Carcharodon hastalis* (formerly in *Isurus* Rafinesque or *Cosmopolitodus* Glickman) to the well-serrated extant white shark, *C. carcharias*, via "semi-serrated" *C. hubbelli* in the Late Miocene, as well as from the Early Paleocene to Early Eocene *O. obliquus* to the species *"megalodon"* via mid–Late Eocene *O. auriculatus,* Early Oligocene *O. angustidens*, and Early–Middle Miocene *O. chubutensis* as a single evolving *Otodus* clade (e.g., Muizon and DeVries, 1985; Ehret et al., 2009a, 2009b, 2012; Pimiento and Balk, 2015; Trif et al., 2016; Shimada et al., 2017, 2021a, 2025; Perez et al., 2019). Although the exact systematic placement of the family Otodontidae remains uncertain (Shimada, 2022; Shimada et al., 2025), in this paper, *"megalodon"* is also considered an otodontid and is placed in the genus *Otodus* (but also see Acknowledgements).

Bendix-Almgreen (1982, p. 28) noted that the associated vertebrae (NHMD 157890) consist of thick, dense concentric "rings" on the articular surfaces, and their inner side (i.e., the corpora calcarea) exhibits rather porous, thin, "stick-like", delicate structures. Whereas the rings which Bendix-Almgreen (1982) referred to most certainly reflect the incremental growth bands, the thin "stick-like" structures are here interpreted to be either the bases of thin radial lamellae (e.g., image of "3121.3" in Cooper et al., 2022, fig. S1) or walls of radial calcified tubes or canals (e.g., Welton, 2013b, fig. 4; Siversson et al., 2019, fig. 16c). Features decisively identifiable as radial lamellae are not preserved in NHMD 157890 consisting mostly of partial or fragmentary vertebrae; however, thin, but solid, radial lamellae are evident in at least some other vertebral specimens of *O. megalodon* in transverse sections or radiographs as well as externally in lateral views (Leriche, 1926, fig. 191; Uyeno and Sakamoto, 1984, plates VI, VII, X; Gottfried et al., 1996, fig. 1C; Kent, 2018, fig. 2.15C). They are consistent with typical lamniform vertebrae (e.g., "primary exochordal radii" as one of the lamniform synapomorphies listed by Compagno, 1990; see also Newbrey et al., 2015, and references therein). Although a more comprehensive survey of the distribution of radial calcified tubes or canals in vertebrae within each individual and species and across different taxa is needed, together with radial lamellae, the presence of a densely spaced radial pattern in NHMD 157890 strongly indicates its placement in the order Lamniformes.

NHMD 157890 co-occurred with associated gill rakers and placoid scales of a cetorhinid (NHMD 1842036), and one may ask whether the associated vertebrae also belong to the basking shark, which is also a lamniform shark. Bendix-Almgreen (1982, 1983) made no mention of the association of the specific large vertebrae (i.e., NHMD 157890) with any basking shark remains, except to note that fossils of basking sharks (e.g., gill rakers and vertebrae) are relatively common in the area and that the diameter-to-length dimension of the associated vertebrae is very different from those of *Cetorhinus*. Although he did not provide any specific data to support these observations, we confirm that cetorhinid remains, particularly gill rak-

ers but also some vertebrae, are relatively common in the Gram Formation (e.g., MSJN 740, formerly MSM P-827; MSJN 1470, formerly MSM P-1433; Figure 4A, B). We also agree with Bendix-Almgreen's (1982, 1983) interpretation that the associated vertebrae (NHMD 157890) are not from a cetorhinid based on vertebral proportion (see Welton, 2015) and apparent lack of strong, sparsely spaced concentric lamellae that characterise Miocene and living cetorhinids (see Welton, 2013a, 2015).

The total vertebral count of extant *Cetorhinus maximus* is only about 109–116, with a precaudal count of 50–54 (Springer and Garrick, 1964; Compagno, 2002). On the other hand, although the total vertebral count in *O. megalodon* remains unknown, the fossil species has at least about 140 vertebrae in the precaudal region alone, based on IRSNB P 9893 (Gottfried et al., 1996; Cooper et al., 2022; Sternes et al., 2024; Shimada et al., 2025). The consequence is that the anteroposterior lengths of cetorhinid vertebrae are generally much longer than those of *O. megalodon* vertebrae. Case in point, the anteroposterior length of vertebrae, particularly larger ones, is generally at least 47% of their diameter in Miocene–Recent cetorhinids (e.g., 47% in MSJN 740 based on Figure 4B; 75% in the 12th vertebra in MCZ 54413 based on Figure 4D; 75–95% in the largest [i.e., within the range of 7th–12th] vertebrae preserved in the following fluid-preserved, "anterior body" specimens of extant *C. maximus* not figured here: ZMUC Jrn9, ZMUC P0571, ZMUC P2401427, and ZMUC P2401428) and may even exceed the diameter (e.g., 104% in CMM-V-861: Gottfried, 1995, p. 445, fig. 1D, E). In contrast, the anteroposterior length of vertebral specimens that have been identified as *O. megalodon* does not exceed 48% of their diameter (e.g., 45% in IRSNB P 9893 based on Figure 4C; ca. 35% in Vertebra "B" described by Uyeno and Sakamoto, 1984, table 1; ca. 25(?)% in the vertebra from South Carolina illustrated by Gottfried et al., 1996, fig. 1C; 48% in CMM-V-4739 illustrated by Kent, 2018, fig. 2.15B). As noted above, Vertebra II is one of the largest vertebrae in NHMD 157890 and measures at least 224 mm in diameter (but likely no more than 230 mm: see above) and 80 mm in anteroposterior length, meaning that its anteroposterior length is at most about 36% of its diameter. This proportion falls within the range of *O. megalodon* (i.e., ≤48%) but outside that of cetorhinids (i.e., ≥47%), which suggests that NHMD 157890 is not from a cetorhinid.

It should be added that, whereas the previously reported largest known fossil cetorhinid vertebrae are only up to 8–9 cm in diameter (e.g., Gottfried, 1995; Welton, 2013b), at least one fossil cetorhinid specimen consisting of 14 associated vertebrae from the Gram Formation (MSJN 740) contains centra measuring almost 16 cm in diameter (Figure 4B). If the large fossil cetorhinid centra are assumed to represent the largest vertebrae in the fossil individual that had a similar ontogenetic growth pattern to extant *C. maximus*, the individual would have measured about 7.5 m TL (e.g., Natanson et al., 2008, fig. 2). Although the fossil record of the family Cetorhinidae goes as far back as the middle Eocene (see Kovalchuk and Barkaszi, 2021; Malyshkina et al., 2022), in fact, there is currently no evidence suggesting that any of the extinct cetorhinids reached gigantic (e.g., >7.5 m TL) proportions (e.g., Hovestadt and Hovestadt-Euler, 2012; Welton, 2013a, 2013b). This contrasts with *O. megalodon* that is known from an 11 m long vertebral column consisting almost entirely of precaudal vertebrae (IRSNB P 9893: see Cooper et al., 2022; Shimada et al., 2025). Even though the variation range and developmental pattern of placoid scales in cetorhinids are yet to be surveyed properly, the size of placoid scales is known to increase ontogenetically at least in some shark species (Raschi and Elsom 1986). However, it is noteworthy that whereas the recovered cetorhinid placoid scales in this study (NHMD 1842036) are morphologically similar to the placoid scales present on the side of the body immediately behind the gill region of a 385 cm TL *C. maximus* (MCZ 54413: Figure 4E), they are about the same size or even smaller than those scales of the extant specimen. Furthermore, although the fossil gill rakers (e.g., Figure 3A) are too fragmentary to estimate the TL of the individual, their U-shaped bight region (Figure 3A, B) is much smaller than that of >6 m TL individuals despite the fact that gill raker size is known to increase ontogenetically (e.g., Hovestadt and Hovestadt-Euler, 2012; Welton, 2013a). Therefore, the sizes of these placoid scales and gill rakers are probably too small to have come from a shark individual with vertebrae measuring up to 23 cm in diameter (NHMD 157890).

One additional line of evidence suggesting that the associated vertebrae (NHMD 157890) likely do not belong to Cetorhinidae is the texture of the corpus calcareum as a function of its weak calcification level. The corpus calcareum of the vertebrae in *O. megalodon* has a dense, solid, well-calcified appearance, including their articular sur-

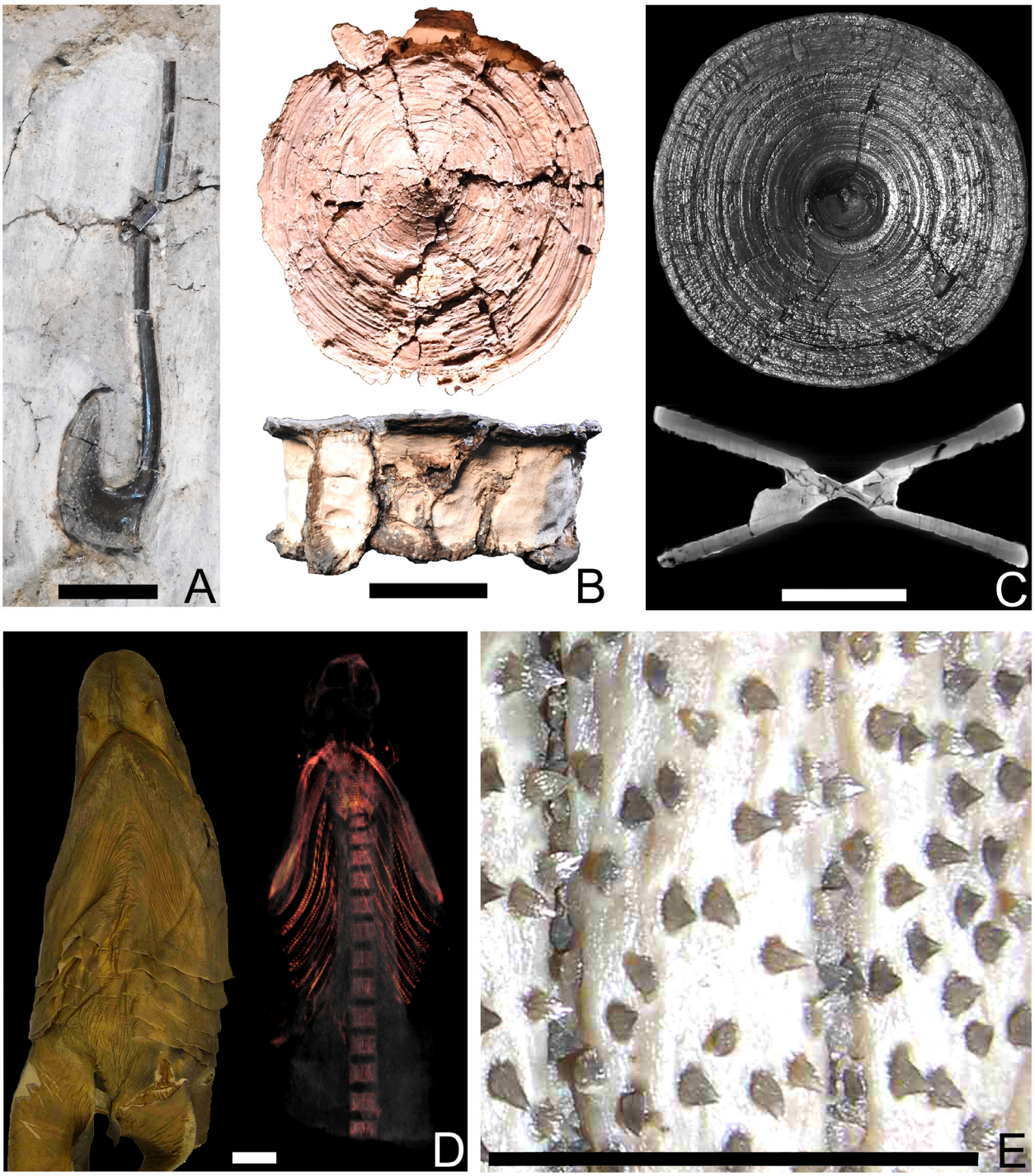


FIGURE 4. Comparative specimens. **A**, One of two fossil cetorhinid gill rakers associated with MSJN 740 (Figure 4B) from the Gram Formation of Denmark (MSJN 1470). **B**, One of the 14 associated fossil cetorhinid vertebrae (MSJN 740), that were also associated with two gill rakers (e.g., Figure 4A) from the Gram Formation of Denmark in articular view (top) and profile view (bottom). **C**, One of the largest associated vertebrae of *Otodus megalodon* from the Miocene of Belgium ("centrum #1" of IRSNB P 9893) in articular view (top: photograph courtesy of IRSNB) and sagittal cross-sectional view (bottom: computed tomographic image). **D**, Preserved "head" specimen of extant *Cetorhinus maximus* (MCZ 54413, 385 cm TL, female, off the coast of Massachusetts, USA) showing ventral external view ((left; photograph) and ventral skeletal view (right; computed tomographic image). **E**, Close-up view of the lateral side of the body immediately above the base of the left pectoral fin of extant *C. maximus* (MCZ 54413, same specimen in "D"), showing sparsely distributed placoid scales (anterior to the left). Bar scales: A, E = 1 cm; B, C = 5 cm; D = 10 cm.

faces that are generally smooth with well-delineated concentric growth bands (e.g., Uyeno and Sakamoto, 1984; Gottfried et al., 1996; Kent, 2018; Cooper et al., 2022). On the other hand, the vertebral corpus calcareum in cetorhinids, including extant *C. maximus*, is relatively thin and poorly calcified, so much so that modern vertebral specimens are often difficult to render even radiographically (Figure 4D) and even become distorted if desiccated (e.g., MNHC F20087, ZMUC P0558, and ZMUC P0578). Fossil vertebrae of Miocene and younger cetorhinids also often exhibit a frail, delicate, poorly calcified appearance (e.g., Gottfried, 1995; Welton, 2013a, 2013b), where the articular surfaces may be rough with poorly delineated growth bands (Figure 4B). Although most specimens are now fragmentary, preserved vertebral parts in NHMD 157890, particularly pieces of corpus calcareum, are quite dense and solid (e.g., Figure 2A, D, H), which is more consistent with the condition of calcification of *O. megalodon* than that of cetorhinids.

In summary, besides the fact that a tooth of *O. megalodon* is known from the same Gram Formation locality (Figure 1G), the associated vertebrae (NHMD 157890) are interpreted to belong to *O. megalodon* based on their exceptional size, short relative length, and robust corpora calcarea, and do not represent parts of a cetorhinid. This means that the remains of the *O. megalodon* individual were intermingled with those of a cetorhinid individual. Given the record size of the largest vertebrae of NHMD 157890 (i.e., Vertebrae I and II), it is reasonable to assume that they originated from the most robust trunk region of the vertebral column, located immediately dorsal to the stomach region. We hypothesize that the cetorhinid materials represent the stomach contents of the *O. megalodon* individual. Because *O. megalodon* occupied a high trophic position with a wide dietary range (e.g., Kast et al., 2022; McCormack et al., 2022, 2025, and references therein), it is not difficult to envision slow-moving cetorhinids as potential prey. The lack of any placoid scales of *O. megalodon* (e.g., see Shimada et al., 2024) in NHMD 157890 is likely not only due to the small amount of sediment sample examined, but also due to taphonomic reasons. Vertebrae from different body positions were found together in a confined area (i.e., larger and smaller vertebrae: see above), which indicates that the shark underwent considerable decay with displacement of the vertebrae by currents or scavengers before burial. Placoid scales of a dead shark lying on the seafloor are known to easily drift away from the carcass (e.g., see Schäfer, 1972, p. 56). Therefore, the placoid scales covering the exterior surface of *O. megalodon* may have been scattered and washed away first, while its internal organ, including the stomach with its contents (i.e., associated cetorhinid gill rakers and placoid scales), remained relatively stationary near or between the *O. megalodon* vertebrae.

Ontogenetic Assessment

Shimada et al. (2021b) examined the growth pattern of *O. megalodon* based on radiographically rendered incremental growth bands assumed to have formed annually in the corpus calcareum of the vertebrae in IRSNB P 9893. The total vertebral band count in the specimen was 46, where the largest vertebral centrum ("vertebra #4") measured 155 mm in diameter, and the individual was estimated to have been 9.2 m TL based on Gottfried et al.'s (1996) work. Subsequently, Shimada et al. (2025) re-analyzed the growth pattern of the individual because its TL was reinterpreted to be 16.4 m TL. That study followed the exact same steps as for Shimada et al.'s (2021b) study in that, first, each growth band was assigned a consecutive band number (BN), where the band at age 0 (= BN 0) was identified by the "angle change" of the corpus calcareum. Next, besides each band interval (BI) from one band to the next between BN 0 to BN 46, the percent centrum radius (pCR) at each growth band was calculated by considering the last band, BN 47, which marked the centrum radius (CR) of 77.5 mm, as 100%. Then, each extrapolated TL (eTL) from each pCR at each BN was calculated by treating the estimated TL for that individual, i.e., 16.4 m or 1,640 cm, as 100%, which also allowed for obtaining an estimated growth length (eGL) gain from one band to the next. The 48 BN-eTL pairs, including eTL at BN 0, were then exploratorily applied to the von Bertalanffy growth function (VBGF) (Bertalanffy, 1938) to fit the BN-eTL data using the least squares method. This quantitative operation involved a hypothetical supposition that each BN-eTL pair, including the eTL at BN 0, comes from a randomly sampled individual of a population, although the BN and eTL values in the study represent dependent measurements from only one individual shark. Although the use of vertebral incremental bands to estimate the age and growth patterns of sharks has limitations, often providing underestimations (e.g., Passerotti et al., 2014; Harry, 2018; Natanson et al., 2018; Natanson and Darcy, 2019; see further discussion below), it is practically the only source

of information offering insights into the growth patterns and life history of extinct elasmobranchs (Shimada, 2008; Newbrey et al., 2015; Amalfitano et al., 2017; Sternes and Shimada, 2018; Shimada and Everhart, 2019; Jambura and Kriwet, 2020; Shimada et al., 2021b, 2025). One peripheral result of Shimada et al.'s (2025) study was the estimated total BN of 83 (i.e., death at age 83) for the *O. megalodon* individual represented by "Bendix-Almgreen's (1983) gigantic vertebrae", assuming: (1) that the extended VBGF curve based on IRSNB P 9893 can be applied to the gigantic vertebrae from Denmark; (2) that the reported diameter of 23 cm is accurate; and (3) that Shimada et al.'s (2025) extrapolation of 24.3 m TL for that individual is reasonable.

The rediscovery of Bendix-Almgreen's (1983) gigantic vertebrae measuring 23 cm in diameter (NHMD 157890) and the recognition of many growth bands, at least radiographically, on parts of Vertebra I (Figure 2B), provided us with an opportunity to investigate the growth pattern of the presumably 24.3 m TL *O. megalodon* individual. As described above, the "angle change" on the corpus calcareum of Vertebra I (and supported by Vertebra II and a natural cast of a vertebral center: Figure 2A, E, F) occurs at about 17 mm away from the vertebral focus, which is denoted by a dot labeled "a" in Figure 2B. This 17 mm position, which is denoted by the arrow "b" in Figure 2A, is considered the vertebral radius at birth, or BN 0. The growth bands between BN 0 and point "c" in Figure 2B are not discernible. However, based on IRSNB P 9893, the growth rate of *O. megalodon* during the first seven or eight years is known to be slightly faster relative to the remaining life of the shark (Shimada et al., 2021b, 2025), where BN 0, 1, 2, 3, 4, 5, 6, and 7 in that specimen are located 16.8, 18.7, 20.5, 22.4, 24.2, 25.8, 27.6, and 29.2 mm away from the vertebral focus, respectively, with an interval range from 1.6–1.9 mm and an average of 1.786 mm (Shimada et al., 2025, table 4). Despite producing an unrealistic linear growth model, we hypothetically applied that average value (i.e., 1.786 mm) to intervals BN1 to BN8 in NHMD 157890 to obtain the centrum radius at each of those eight bands. Because BN 0 in NHMD 157890 is 17.0 mm from the vertebral focus, the position of BN 1 through BN 8 in NHMD 157890 would be at 18.8, 20.6, 22.4, 24.1, 25.9, 27.7, 29.5, and 31.3 mm from the vertebral focus, respectively. The asterisk (*) in Figure 2B indicates the assumed approximate position of BN 8. Because the band intervals beyond BN 8 are relatively consistent in IRSNB P 9893, we assume it to be also the case in NHMD 157890, as is also indicated by the band intervals observed between "c" and "d". Because the assumed position of BN 8 in this study and the measured position of "c" are, respectively, 31.3 mm and 80.9 mm from the vertebral focus, our calculation suggests that 29 growth bands would fit in the 49.6 mm interval between the two points if the average interval of 1.706 mm between "c" and "d" is used even though an unrealistic linear growth model needs to be assumed (see Description above). With these assumptions, every observed or extrapolated growth band in Vertebra I is assigned a band number (Figure 5A), including "c", "d", and "e" (Figure 2B, C), which are determined to be BN 37, 53, and 64, respectively.

Based on the total of 65 growth bands (including BN 0), CR at each growth band was directly measured or calculated. Moreover, BI, pCR, eGL, and eTL at each respective growth band were calculated using Shimada et al.'s (2021b, 2025) method, except for considering the TL of 2,430 cm at BN 64 as 100% to back-calculate the eTL at each previous BN (Table 1). Likewise, Shimada et al.'s (2021b, 2025) methods and assumptions were used for our VBGF analysis based on the 65 BN-eTL pairs, including the computations of growth parameters using the Desmos Inc. graphing software (www.desmos.com, API v1.12) with the following VBGF formula describing the length (L) as a function of the age of the shark (t):

$$L_{(t)} = L_{\infty}\,(1 - e^{-k(t - t_0)})$$

where L_{∞} is the estimate of asymptotic (= maximum) length, k the rate constant with units of reciprocal time (i.e., the duration for a fish in a population to attain near its mean maximum length), and t_0 the theoretical time at zero length. In addition, the y-intercept of the attained VBGF curve represents the body length at birth (L_0). We also tentatively estimated the longevity of the shark using Natanson et al.'s (2006) equation for the estimated age at 95% of L_{∞}, i.e.:

$$Longevity = (1/k)ln\{(L_{\infty} - L_0)/[L_{\infty}(1 - x)]\} \text{ with } x = L_{(t)}/L_{\infty} = 0.95$$

Figure 5B shows the VBGF fitted to correlate 65 BN values with their corresponding TL values (Table 1). The VBGF parameters of the non-linear regression line (r^2 = 0.998) are L_0 = 360.000 cm TL, L_{∞} = 7,064.765 cm TL, and k = 0.00693 yr^{-1}, and the theoretical longevity of 424.738 years. However, this VBGF curve is considered highly unreliable even though its coefficient of determina-

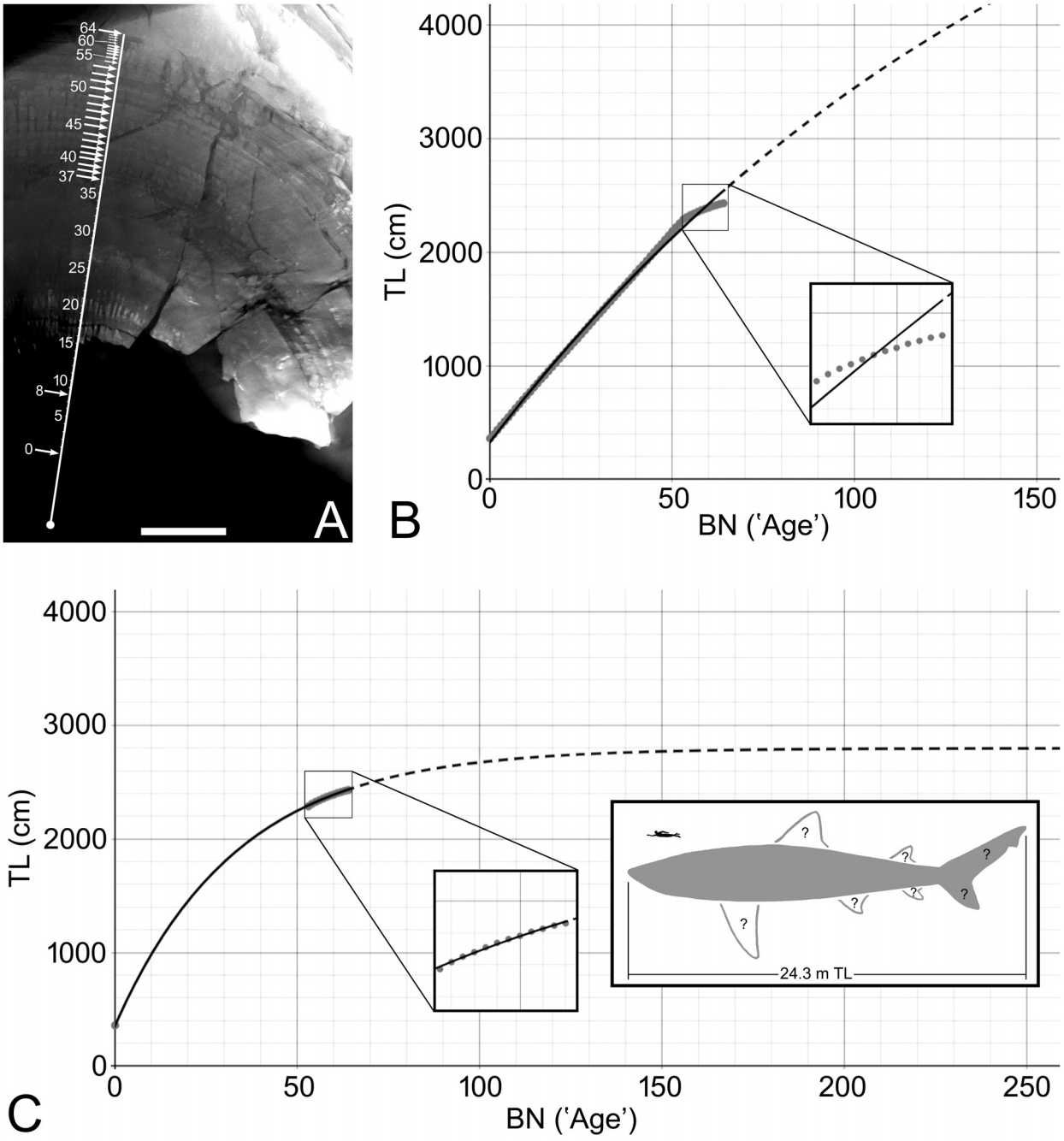


FIGURE 5. Ontogenetic growth analyses of *Otodus megalodon* based on NHMD 157890 (see text for details). **A**, The same microCT image as in Figure 2B but showing observed and hypothetically assumed growth bands or "band numbers" (BN) (bar scale equals 2 cm). **B**, von Bertalanffy growth function (VBGF; curved line) fitted to all 65 data points (including BN 0; Table 1) depicting the relationship between growth band numbers (BN, or "age", of the individual in years) and total lengths (TL), along with inset showing a close-up view of the last 12 plots that poorly fit to the VBGF curve. **C**, The same VBGF (curved line) as Figure 5B but fitted only to BN 0 and the last 12 growth bands (BN53–BN64), along with the left inset showing a close-up view of the last 12 plots that fit well with the VBGF curve, and the right inset showing Shimada et al.'s (2025, fig. 5C) highly tentative reconstruction of *O. megalodon* with a tentative TL of 24.3 m TL based on NHMD 157890, along with silhouette of *Homo sapiens* illustrating the relative length of the reconstructed *O. megalodon* for comparisons (Note: It must be emphasized that *H. sapiens* and *O. megalodon* never coexisted and that the size and shape of all the fins, including the caudal fin, remain highly hypothetical).

TABLE 1. Raw and derived measurements from Vertebra I in NHMD 157890 (Figures 2A–C, 5A) inferred to be from a 24.3 m TL individual, including its estimated TL at each growth band (eTL-1), along with comparisons with the estimated TL at each growth band in IRSNB P 9893 estimated to be 16.4 m TL with up to only 46 growth bands (eTL-2) described by Shimada et al. (2025). Abbreviations: BN, band number; CR, centrum radius; BI, band interval from the previous band; pCR, percent centrum radius from the center of the vertebra; eTL, extrapolated total length of entire shark; eGL, estimated growth length gain from the previous year. Asterisks(*) indicate hypothetical CR and BI values used for the ontogenetic growth analysis (Figure 5; see text for details).

BN	CR (mm)	BI (mm)	pCR (%)	eGL (cm)	eTL-1 (cm)	eTL-2 (cm)
0	17.0	-	14.8	-	360	356
1	18.8*	1.786*	16.3	36	396	395
2	20.6*	1.786*	17.9	39	435	433
3	22.4*	1.786*	19.5	39	474	474
4	24.1*	1.786*	21.0	36	510	512
5	25.9*	1.786*	22.5	37	547	546
6	27.7*	1.786*	24.1	39	586	584
7	29.5*	1.786*	25.7	39	625	618
8	31.3*	1.8*	27.2	36	661	649
9	33.1*	1.706*	28.8	39	700	677
10	34.8*	1.706*	30.3	36	736	705
11	36.5*	1.706*	31.7	34	770	731
12	38.3*	1.706*	33.3	39	809	754
13	40.0*	1.706*	34.8	36	845	777
14	41.7*	1.706*	36.3	37	882	802
15	43.4*	1.706*	37.7	34	916	827
16	45.1*	1.706*	39.2	37	953	848
17	46.8*	1.706*	40.7	36	989	872
18	48.5*	1.706*	42.2	36	1025	902
19	50.2*	1.706*	43.7	37	1062	930
20	51.9*	1.706*	45.1	34	1096	956
21	53.6*	1.706*	46.6	36	1132	984
22	55.3*	1.706*	48.1	37	1169	1012
23	57.0*	1.706*	49.6	36	1205	1038
24	58.7*	1.706*	51.0	34	1239	1069
25	60.4*	1.706*	52.5	37	1276	1099
26	62.1*	1.706*	54.0	36	1312	1130
27	63.8*	1.706*	55.5	37	1349	1159
28	65.5*	1.706*	57.0	36	1385	1187
29	67.3*	1.706*	58.5	37	1422	1217
30	69.0*	1.706*	60.0	36	1458	1246
31	70.7*	1.706*	61.5	36	1494	1268
32	72.4*	1.706*	63.0	37	1531	1291
33	74.1*	1.706*	64.4	34	1565	1312
34	75.8*	1.706*	65.9	36	1601	1335
35	77.5*	1.706*	67.4	37	1638	1361
36	79.2*	1.706*	68.9	36	1674	1391
37	80.9	1.706*	70.3	34	1708	1414
38	82.6	1.7	71.8	37	1745	1433
39	84.2	1.6	73.2	34	1779	1463
40	86.0	1.8	74.8	39	1818	1489
41	87.8	1.8	76.3	36	1854	1515
42	89.4	1.6	77.7	34	1888	1542
43	91.1	1.7	79.2	37	1925	1566
44	92.9	1.8	80.8	38	1963	1589
45	94.7	1.8	82.3	37	2000	1614
46	96.4	1.7	83.8	36	2036	1640
47	98.0	1.6	85.2	34	2070	
48	99.8	1.8	86.8	39	2109	
49	101.6	1.8	88.3	37	2146	
50	103.4	1.8	89.9	39	2185	
51	105.0	1.6	91.3	34	2219	
52	106.6	1.6	92.7	34	2253	
53	108.2	1.6	94.1	34	2287	
54	109.3	1.1	95.0	22	2309	
55	110.1	0.8	95.7	17	2326	
56	110.8	0.7	96.3	14	2340	
57	111.4	0.6	96.9	15	2355	
58	112.1	0.7	97.5	14	2369	
59	112.7	0.6	98.0	12	2381	
60	113.2	0.5	98.4	10	2391	
61	113.7	0.5	98.9	13	2404	
62	114.2	0.5	99.3	9	2413	
63	114.6	0.4	99.7	10	2423	
64	115.0	0.4	100	7	2430	

tion (r^2 = 0.998) is high, particularly because the fitting of the plots to the curve is poor around the last 20 plots (e.g., inset of Figure 5B, which shows the last 12 plots). The poor fit is almost certainly affected in part by the fact that the growth from BN 0 (= “b” in Figure 2B) to BN 8 (= “*” in Figure 2B), as well as that from BN 8 to BN 37 (= “c” in Figure 2B), were artificially constructed as linear (see above). In addition, a rather tight bending of the curve around BN53, which marks “d” in Figure 2B, suggests that the observed growth band even

between "c" and "d" may be inaccurate, including the possibility that the total growth band number of 64 may be a conservative undercount.

Based on the possible factors that may explain the poor fit (Figure 5B) with an impossibly high L_∞ of 70 odd meters, we therefore conducted another VBGF analysis using just the first plot (i.e., BN 0 with an extrapolated TL at birth of 360 cm) and the last 12 plots (i.e., comprising only the observed growth bands from "d" in Figure 2C: i.e., BN 53–64). Figure 5C shows the resulting non-linear regression line with the VBGF parameters of L_0 = 360.000 cm TL, L_∞ = 2,798.411 cm TL, and k = 0.02978 yr^{-1}, with a theoretical longevity of 95.971 years. The regression line has a very high coefficient of determination (r^2 = >99.9%), but more importantly, the 12 plots have constrained the curved line to align closely with their positions (i.e., inset of Figure 5C, which shows the last 12 plots). Although we regard the regression line in Figure 5C and its VBGF parameters to be more reasonable (see further discussion below), because of the possibility of band number undercount, we conducted an additional analysis with the same 360 cm TL at BN 0 but hypothetically considering those last 12 growth bands to represent the last 12 bands of a vertebra with a total growth band count of 83 (i.e., BN 72–83), which was predicted for Bendix-Almgreen's (1982, 1983) gigantic vertebrae by Shimada et al. (2025, fig. 6B). Although the VBGF curve is not graphically illustrated here because of many hypothetical suppositions, the VBGF parameters of that non-linear regression line (r^2 = >0.999) are L_0 = 360.000 cm TL, L_∞ = 3,127.111 cm TL, and k = 0.01661 yr^{-1}, with a theoretical longevity of 172.994 years.

It is important to note that the TL of 2,430 cm for NHMD 157890 with a vertebral diameter of 230 mm was extrapolated from the estimated 1,640-cm-TL *O. megalodon* represented by IRSNB P 9893 with a vertebral diameter of 155 mm (Shimada et al., 2025). Therefore, there is no surprise that the back-calculated size at birth in this study (i.e., "pre-determined" L_0 of 360 cm TL) is very close to that attained in Shimada et al.'s (2025) study (i.e., 356 cm TL). Although the assumption that the two likely non-contemporaneous *O. megalodon* individuals, NHMD 157890 and IRSNB P 9893, had an identical growth pattern is rather unrealistic, it is noteworthy that the TL of 1,640 cm, which was assumed as the TL at the last growth band (BN 47) in IRSNB P 9893, occurs around BN 35 in NHMD 157890. If this discrepancy is taken at face value, our total band count of 64 may be short of a dozen or so bands, which would mean that NHMD 157890 (at least Vertebra I) could have had as many as around 76 growth bands. This in turn would suggest that the VBGF curve and each respective associated parameter could have lied somewhere between the growth curve shown in Figure 5C with up to BN 64 and that (not figured but see above) with up to BN 83. However, because all three models (i.e., BN 64, BN 76, and BN 83 cases) are highly theoretical with many assumptions, including the TL of 2,430 cm used based on IRSNB P 9893), we consider the most conservative growth curve model among the three, i.e., the one shown in Figure 5C, to be currently the most sensible ontogenetic growth model for *O. megalodon*, or at least for the specific individual represented by NHMD 157890. Therefore, we contend that the size at birth for *O. megalodon* was around 3.6 m TL (see also Shimada et al., 2025), the theoretical maximum attainable size was no more than 28 m TL, and the theoretical longevity was no more than 96 years.

Comments on the Size of *O. megalodon*

Otodus megalodon was at least partially endothermic (Griffiths et al., 2023) and most likely possessed a slenderer body than *C. carcharias* based on IRSNB P 9893, which came from an individual estimated to be 16.4 m TL (Sternes et al., 2024; Shimada et al., 2025; Figure 5C, right inset). However, it must be noted that IRSNB P 9893, which was used to estimate the TL of 24.3 m TL for NHMD 157890 examined here, is an incomplete specimen, comprising part of the vertebral column. The TL estimate for IRSNB P 9893 was based on a combination of actual measurements and multiple assumptions, including that: (1) the 11 m long preserved vertebral column represents the entire precaudal trunk region; (2) the estimated head and caudal fin lengths are based on relevant extant shark taxa and provide a reasonable template for estimating TL for IRSNB P 9893; and (3) the quantitative relationship between the TL and vertebral diameter is isometric (Shimada et al., 2025). The reality is that even the estimated total length of 16.4 m for IRSNB P 9893 (Shimada et al., 2025) remains a working hypothesis, although the margin of error is capped at the lower end by the actual length of the preserved part of the vertebral column. Whether the vertebral proportion in *O. megalodon* remained isometric throughout ontogeny remains to be substantiated (e.g., Gayford et al., 2024), as different types of species-specific allom-

etry are known to occur in sharks (e.g., Carbonara et al., 2025, and references therein).

To further complicate the body size estimations of *O. megalodon*, whether the two individuals represented by IRSNB P 9893 and NHMD 157890 are contemporaneous cannot be ascertained because the exact stratigraphic provenance of IRSNB P 9893 within the Antwerp region in Belgium is uncertain (see Gottfried et al., 1996). Moreover, it is possible that the growth pattern and the maximum size range of *O. megalodon* varied from local population to local population. For example, Bergmann's rule may have acted on the species for gigantism in higher latitude (or cooler water) populations relative to lower latitude (or warmer water) populations (Shimada et al., 2022; but see also Herraiz et al.'s (2026) work, which is questionable because, besides unjustifiably claiming Shimada et al.'s (2022) "already-refuted" nurseries of *O. megalodon* based on finer time bins and consideration of paleolatitudes and sea-surface paleotemperatures as "well-documented nurseries", their study is based on an incoherent mixture of latitudinally and geographically confined specimens from multiple unevenly-sized depositional basins with practically no stratigraphic or taphonomic context, over half of which are lower anterior teeth known to yield unreliable TL estimates (see Shimada, 2019; Perez et al., 2021) and include significantly damaged teeth with unverifiable tooth measurements and tooth position assignments). Although the latitudinal difference between Antwerp, Belgium, and Gram, Denmark, is only about 4°, the locality of NHMD 157890, which also yielded MSJN 1540 (Figure 1G), nevertheless, marks (paleo)latitudinally the highest occurrence record for *O. megalodon* worldwide substantiated by cataloged specimens (55.5° North: Pimiento et al., 2016, supplementary information; see also Bateman and Larsson, 2025, 2026). In addition, it is noteworthy that, whereas NHMD 157890 clearly comes from a gigantic individual, the isolated tooth (MSJN 1540: Figure 1G) is likewise from a large individual. The tooth is likely an upper anterior tooth (or one of the larger mesially located upper lateral teeth) with an estimated crown height of about 113 mm, which would mean that it came from an individual that minimally measured 13.3 m TL based on Shimada's (2019, table 1, "U") equation known to yield significant underestimations for this species (Sternes et al., 2024; Shimada et al., 2025). In fact, *O. megalodon* individuals estimated to be 13+ m TL based on the same TL extrapolation method are rather rare globally (e.g., see Shimada et al., 2022, fig. 2a).

The use of VBGF to investigate the age and growth patterns of elasmobranchs is common, particularly in the context of their conservation biology and management decisions, where a large sample size is ideally needed to adequately capture the local population structure (e.g., Cailliet and Goldman, 2004; Goldman et al., 2012; Harry, 2018; Fadool et al., 2024, and references therein). Therefore, it must be emphasized that a "single-sample VBGF analysis" employed here using Vertebra I in NHMD 157890 and similar previous studies on other taxa (e.g., Shimada, 2008; Newbrey et al., 2015; Jacobs and Shimada, 2018; Shimada and Everhart, 2019; Shimada et al., 2021b, 2025) are purely "experimental", only capturing a possible ontogenetic growth pattern of that specific individual. Our age estimates of NHMD 157890 rely on the assumption that the concentric growth bands formed annually. However, taxon-specific as well as regionally dependent exceptions in growth band formation do exist in extant shark species, and observed growth bands may not necessarily record age or time but merely indicate "growth" (Passerotti et al., 2014; Harry, 2018; Natanson et al., 2018; Natanson and Darcy, 2019). Therefore, the results of the VBGF presented here, with an estimated theoretically maximum possible TL of almost 28 m with a total growth band count of 64 (Figure 5B) or even nearly 31 m with a hypothetical growth band count of 83 (see text above) for *O. megalodon*, must be viewed as tentative with a large margin of error. The perceived growth pattern gleaned from NHMD 157890 (Figure 5) applies to a single individual and not the entire species. Although previous maximum TL estimates of 15–20 m for *O. megalodon* based on dental or vertebral sizes of the extant *Carcharodon carcharias* (e.g., Shimada, 2019; Perez et al., 2021) are regarded as underestimations (Sternes et al., 2024; Shimada et al., 2025), we nevertheless conservatively contend the gigantic individual represented by NHMD 157890 to be likely one of the rarest size-wise outliers for the fossil species. In fact, we strongly suggest retaining "24.3 m TL" as presently the most reasonable estimated maximum size scientifically for *O. megalodon*. Ultimately, at least one complete or nearly complete skeleton of *O. megalodon* in association with its teeth is needed for any further discussion on the size of the fossil species.

The rediscovery of Bendix-Almgreen's (1982, 1983) vertebral specimen (NHMD 157890; Figure 2) is significant because the currently largest possi-

ble reasonable TL estimate of 24.3 m for *O. megalodon* (Shimada et al., 2025) no longer relies on mere photographic evidence. In a broader biological context, NHMD 157890 is even more unique because it possibly represents the largest vertebral specimen of a fish (= non-tetrapod vertebrate) recorded to date. In fact, while the vertebrae of many gigantic fishes (e.g., dunkleosteid placoderms and suspension-feeding pachycormiform actinopterygian: Liston et al., 2013; Engelman, 2024) are unossified and thus not preserved in the fossil record, the largest recorded calcified vertebrae of chondrichthyans (notably sharks) in the fossil record measure no more than about 16 cm in diameter (e.g., Early Cretaceous lamniforms: Shimada, 1997; Frederickson et al., 2015; Bazzi et al., 2025). Although the maximum vertebral size of the largest extant form, the whale shark [*Rhincodon typus* (Smith)] with the largest recorded individual of 18.8 m TL (McClain et al., 2015), is unknown, it most likely does not exceed the size of vertebrae represented by NHMD 157890, given that the largest recorded extant whale shark vertebra is only about 10 cm in diameter that came from a 10 m TL individual (see Ong et al., 2020, fig. 1, table 1). To our knowledge, the largest reliably measured vertebra of the second largest extant fish, *Cetorhinus maximus*, is nearly 17 cm in "lateral diameter" from an 8.5 m TL individual (Natanson et al., 2008, fig. 2). Because the largest reliably measured *C. maximus* on record is 12.27 m TL (McClain et al., 2015), that individual could have had vertebrae measuring about 24 cm in diameter if the ontogenetic relationship between the TL and vertebral diameter is assumed to be isometric. However, whereas there is a possible negative allometry of vertebral diameters in *C. maximus* (Natanson et al., 2008, fig. 2), the physical evidence of such gigantic vertebral specimens of the basking shark or any other extant fish taxa is not known, making NHMD 157890 (particularly Vertebrae I and II: Figure 2), the empirically largest fish vertebrae ever recorded.

CONCLUSIONS

NHMD 157890 is an associated set of disarticulated vertebrae of *O. megalodon* from the Upper Miocene Gram Formation in Gram, Denmark. The specimen was originally described and illustrated by Bendix-Almgreen (1982, 1983), but its whereabouts subsequently became unknown. The rediscovery of the specimen led to this present study, which not only confirmed many of Bendix-Almgreen's (1982, 1983) observations, including the maximum vertebral diameter of 23 cm, but also the vertebrae's co-occurrence with associated gill rakers and placoid scales of a fossil cetorhinid. Because the morphology and diameter-to-length relationships of vertebrae in NHMD 157890 are more consistent with those of *O. megalodon* than cetorhinids, the vertebrae are determined not to be part of the associated cetorhinid remains, alluding to the possibility that the basking shark remains may represent gastric residues of *O. megalodon*. Observable growth bands on Vertebra I of NHMD 157890 were used to decipher its ontogenetic growth pattern, indicating a size at birth of about 3.6 m TL and a theoretical maximum attainable TL of 28 m with a theoretical longevity of 96 years old. As these size and age estimates are based on incomplete remains of *O. megalodon* and several major assumptions, they must be regarded as highly tentative.

ACKNOWLEDGMENTS

A very special thanks goes to F. Osbæck for helping to save the remains of the broken shark vertebrae back in 1989, to B.E. Kramer Lindow for locating and recognizing in 2017 the origin and value of the fragments of the shark vertebrae in the collection of NHMD that are now represented as NHMD 157890, and to L. Rasmussen (MSJN) for patiently attempting to reassemble fragments of the broken pieces of the specimen. We also thank the following individuals for loaning specimens or supplying CT data or specimen information used for this study: A. Folie, S. Beaudart, C. Cousin, J. Lalanne, and U. Lefèvre (IRSNB); K.E. Hartel and A. Williston (MCZ); C.C. Fernandes and K. Doyle (MNHC), and P.R. Møller (ZMUC). We also thank W. Schwarzhans for the identification of otoliths illustrated in Figure 3C, as well as H.M. Maisch, V.J. Perez, M.A. Becker, P.C. Sternes, M.L. Griffiths, and J.H. Gayford for useful discussions. The two anonymous reviewers' constructive comments and suggestions greatly improved the quality of this paper. In addition, we note that not everyone (MS) considers the megatooth species to belong to the genus *Otodus* but to another otodontid genus *Carcharocles*, but this generic interpretation difference does not affect the content expressed in this paper.

DATA AVAILABILITY

MicroCT data of vertebrae I and II of NHMD 157890 is available on MorphoSource: https://www.morphosource.org/, in full resolution datasets: media# 000803915 (36.6 GB, Vertebra I, scan 1),

media# 000804071 (34.1 GB, Vertebra I, scan 2) and media# 000804594 (32.5 GB, Vertebra II) and more manageable 3× down-sampled versions: media# 000804816 (1.3 GB, Vertebra I, scan 1, 3× down-sampled), media# 000804856 (1.3 GB, Vertebra I, scan 2, 3× down-sampled) and media# 000804868 (1.2 GB, Vertebra II, 3× down-sampled).

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